



MICHIGAN STATE
UNIVERSITY



Astronomy for non-STEM majors: Does teaching with a focus on science skills improve student learning outcomes?

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Motivation



- ❖ A diverse group of students take science classes for all kinds of reasons.
- ❖ Not all students will be excited by the subject material. So how do we motivate these students?
- ❖ My project focused on teaching the scientific method and science skills with the hope that students will use them in their own lives

Research Questions

I revised 4 intro astronomy labs and taught them using a Problem Based Approach with a focus on developing scientific skills:

1. How does this affect student's beliefs and feelings about astronomy and its relevance to them?
2. How does this affect their performance with the material?

The Lab: ISP 205L Visions of the Universe

Intro Astronomy Lab

Covers a wide range of
astronomy topics

Students: Non-STEM majors

All academic years

Course requirement

Section 1 - revised labs

Section 2 - control group

The Earth's Seasons

Phases of the Moon

Gravity and Orbits

Imaging an Exploding Star

Stellar Spectra

Exoplanets Wobbles

Habitable Zones

Galaxy Morphology

Objectives

1. Determine student's current feelings about astronomy, science, and lab work and its relevance to them.

2. Compare and contrast student's survey responses prior to and after being taught with the revised lessons.

3. Assess and evaluate student's lab grades in sections with and without the revised labs. (Do they perform better?)

4. Assess student reflections on the scientific skills they learned in lab.

Methods

Section 1

1/26 Pre Survey
2/02 Revised Lab 2 🌑
2/09 Revised Lab 3 🌩️
2/16 Revised Lab 4 🪐
2/23 Revised Lab 5 💥
2/23 Post Survey

3/1 Analysis
Reflection Responses Visualized
Grade Distributions Compared
(2 sample T-test)
Pre and Post Surveys Compared

Section 2

1/27 Pre Survey
No Lab Revisions
2/24 Post Survey
3/1 Analysis
Grade Distributions Compared
Pre and Post Surveys Compare

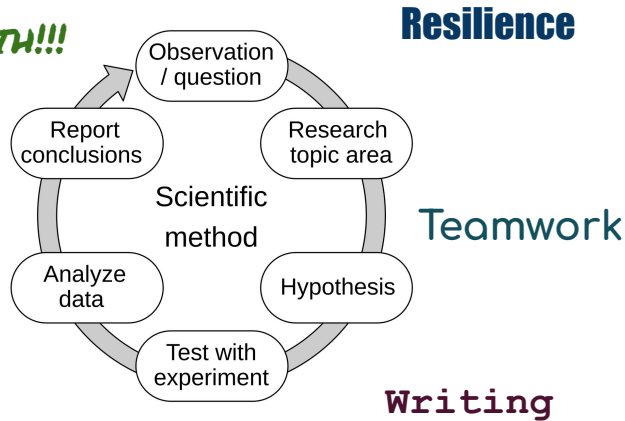


Problem Based Approach

- ❖ Concept overview
- ❖ Real world problems
- ❖ Group based problem solving
- ❖ Structured reflection
- ❖ Emphasis on scientific method
- ❖ Reflection questions ask students what science skills they learned in the lab and how they might use those skills outside of the lab.

Smith et al. 2022

MATH!!!



What's Missing From This Solar Eclipse? 🌞🌑

Have you ever stopped to think that when you see a solar eclipse, you never see the moon before or after the eclipse?

Most people don't. What we see is a black circle appearing out of nowhere and moving across the face of the sun.

The point being, a solar eclipse is not what they taught you in school. Nobody knows exactly what it is, but without a doubt, it's not the moon blocking out the sun.

Discussion:

Why do we see the moon? (What are we seeing? Think about where the light comes from) .

What phase is the moon in during a solar eclipse? What side of the moon is facing us (light or dark)?

When the moon is full (at its brightest) certain stars are less visible. How would the brightness of the sun affect the moon's visibility?



Astronomy And Science Student Attitudes (ASSA): A Short Review And Validation Of A New Instrument

Sophie Bartlett, Cardiff University, United Kingdom

Michael T. Fitzgerald, Edith Cowan University, Australia

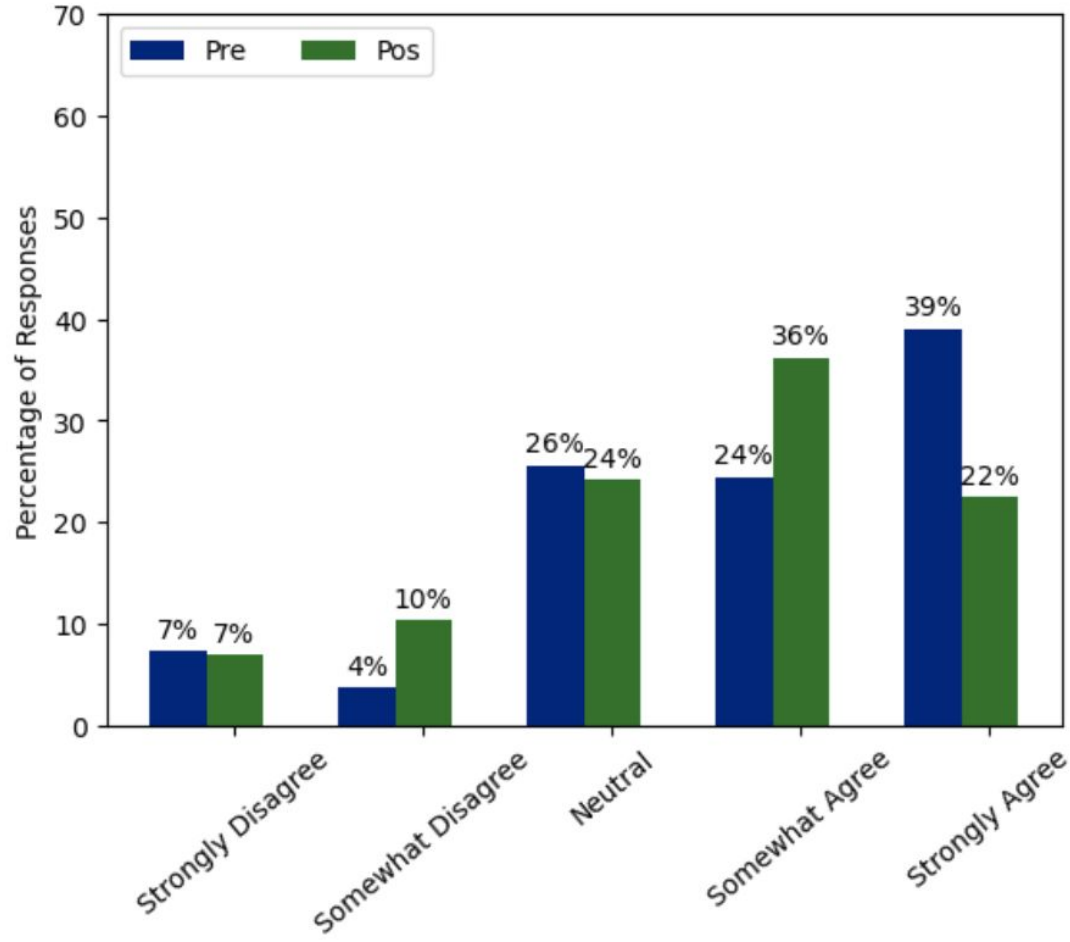
David H. McKinnon, Edith Cowan University, Australia

Lena Danaia, Charles Sturt University, Australia

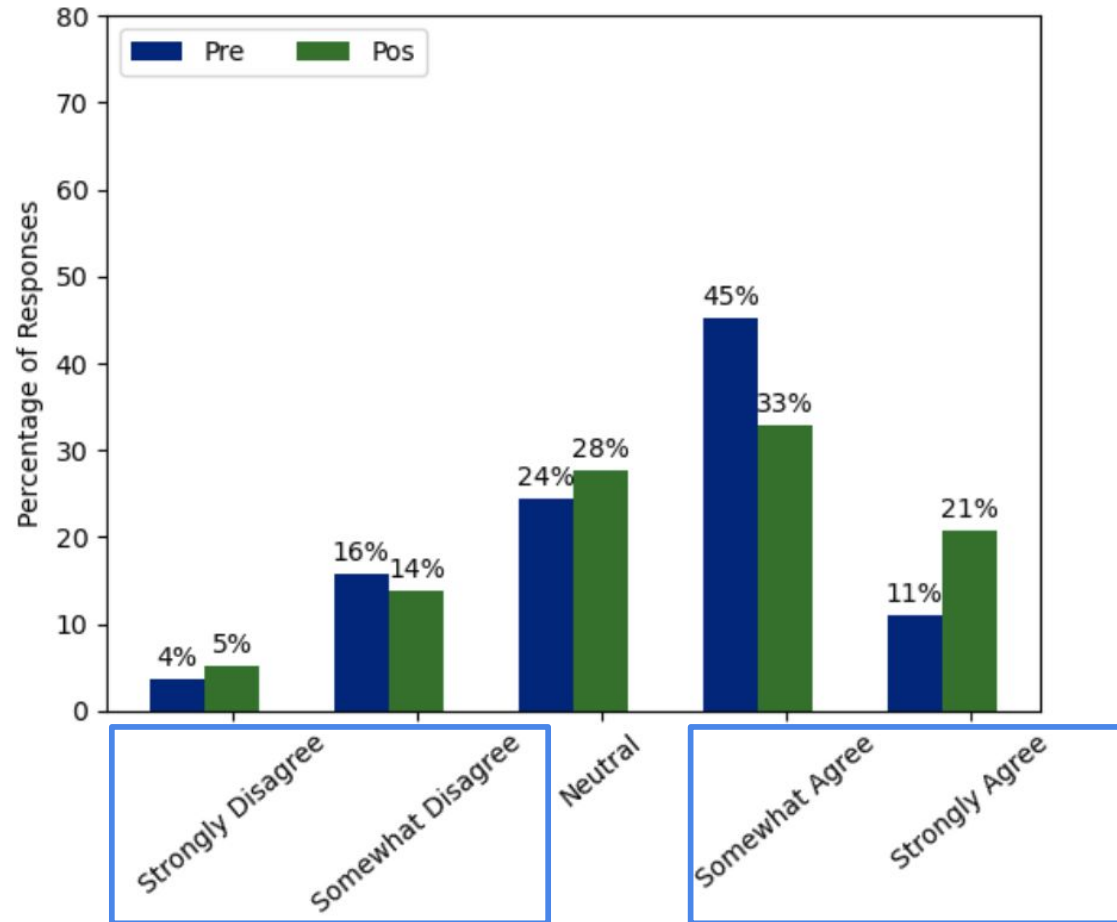
Jasmina Lazendic-Galloway, Monash University, Australia

Section 1 Survey Results

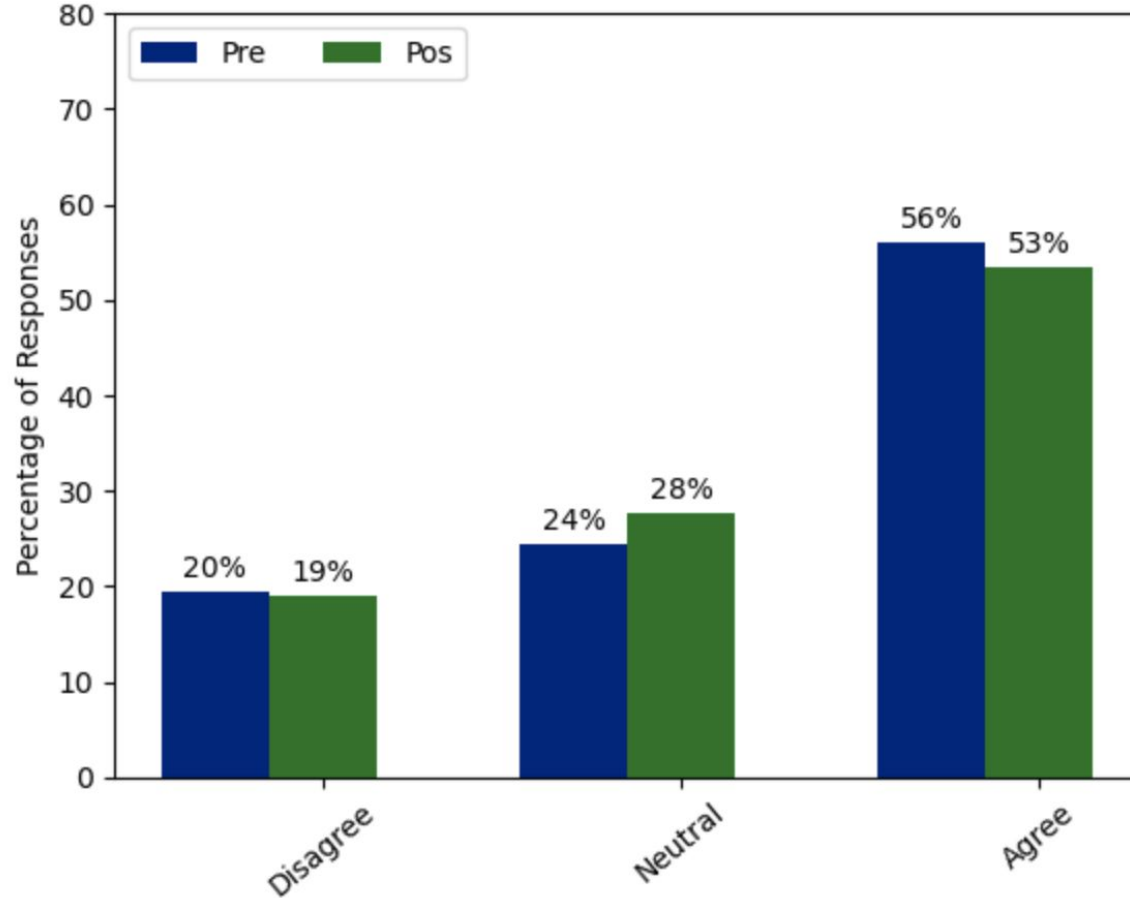
I Would Rather Do An Astronomy Experiment Than Read About Astronomy.



I Find Science Difficult.

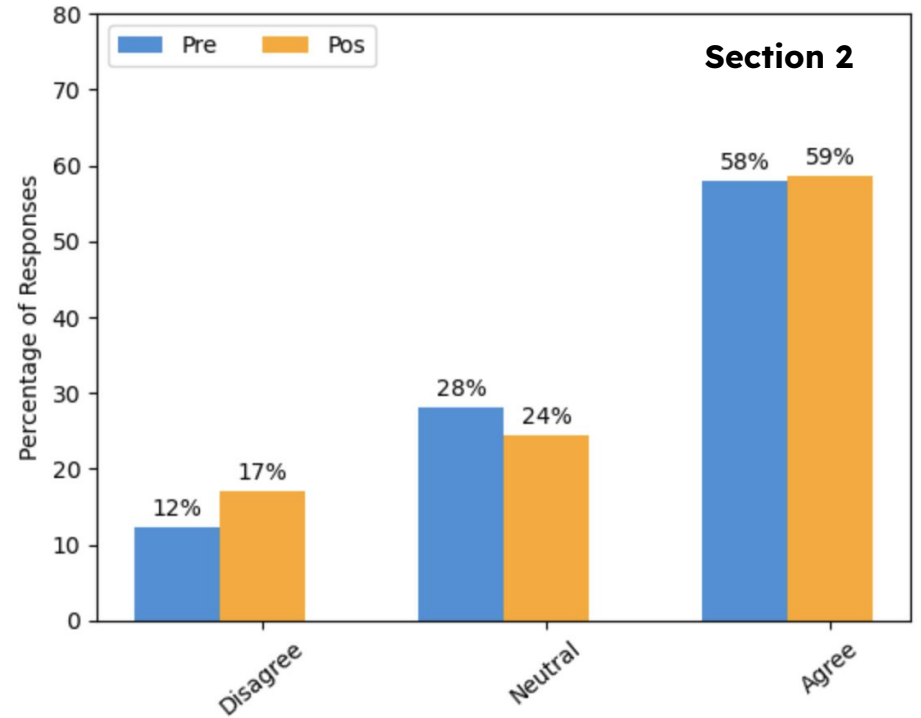
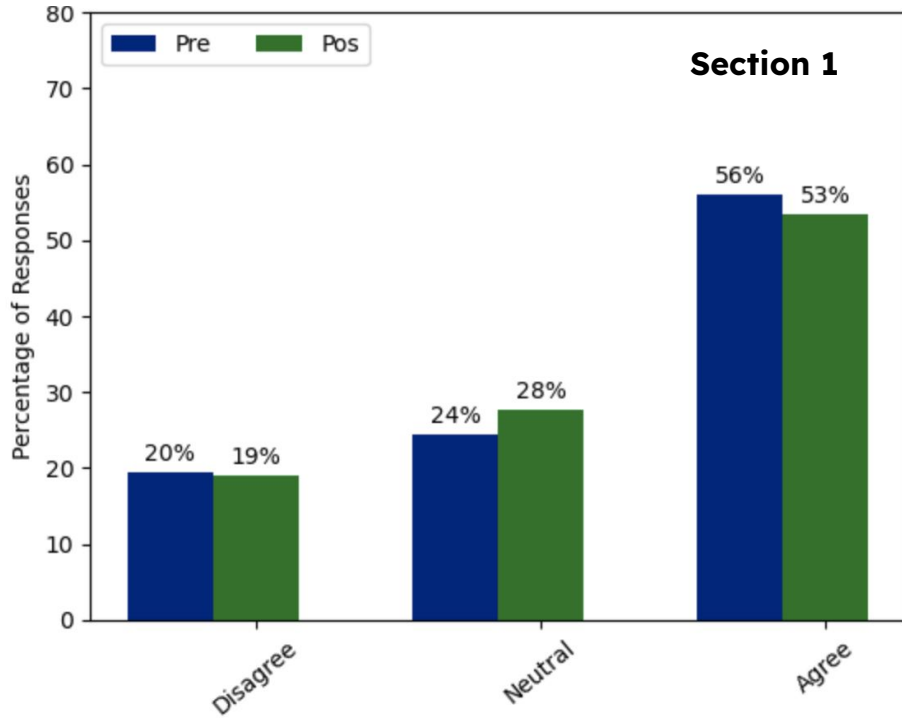


I Find Science Difficult.

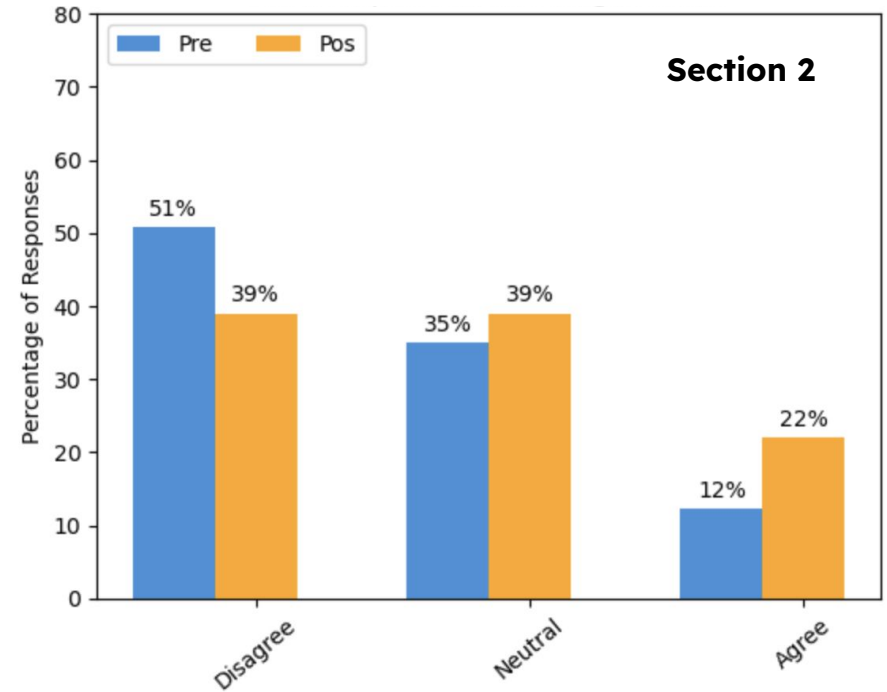
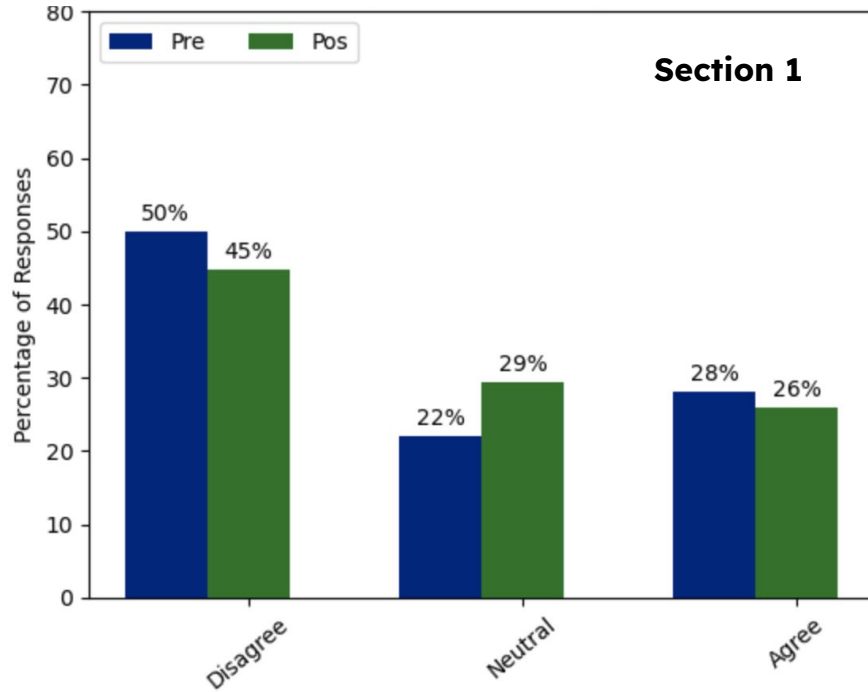


Comparing Section 1 & 2 Survey Results

I Find Science Difficult.



I Feel Helpless When Doing Science.



Grade Distributions

Lab 2

Lab 3

Lab 4

Lab 5

Sec. 1
Avg. = 82.93

Sec. 1
Avg. = 79.73

Sec. 1
Avg. = 78.02

Sec. 1
Avg. = 80.05

Sec. 2
Avg. = 83.03

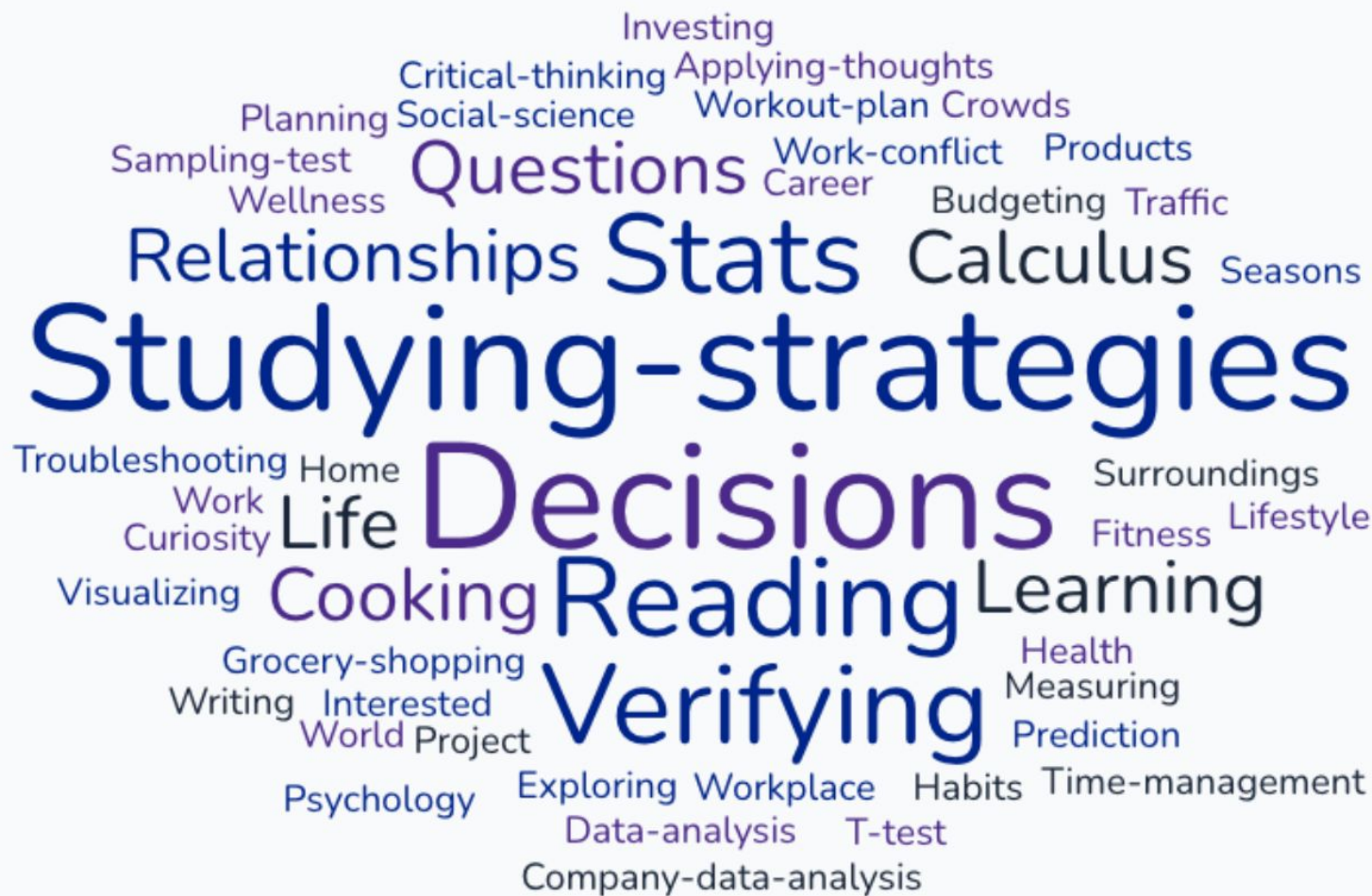
Sec. 2
Avg. = 83.24

Sec. 2
Avg. = 79.05

Sec. 2
Avg. = 82.16

Reflection Questions

Product-demo
Teaching Art Economics
Literature Predictions
Markets Taxes Planets Legos Marketing
Teamwork Career Math Building
Budgeting Skin-care
Business Sports Visualizations Stocks
Finance Stats Testing Engineering
Speed Comparing Classes
Children Disagreements
2D-3D-models



Conclusions

Survey Results

There is a positive change in student attitudes around science.

Grade Distributions

There is no statistically significant difference between sections 1 and 2

Reflection Responses

Student responses were linked to their own interests and lives. Some responses were generic

Lessons Learned

Reflections/Discussion

- ❖ Have time set aside just for reflection and discussion
- ❖ Be specific about what I want students to reflect on. If it's too open ended some students will just give generic responses
- ❖ Use P.A.R.
Problem
Action
Result

Lessons Learned

Class Observations

- ❖ A simple pre class math problem may help students feel a little more comfortable or get them ready for the math they need for class that day.
- ❖ Less is more, keep class work to a manageable length.

Implications

Future Work

- ❖ Apply this framework to an entire class (will keep things consistent)
- ❖ Use more detailed reflection questions that push students to apply their knowledge and skills to their lives
- ❖ Use the reflection questions to make problems.



Thank You!

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